

Answer all questions in this section.

- 1 The information provided in the insert is taken from a published scientific article. Other published articles may not agree with all of this information.

You should read the whole insert before you start to answer any questions and use the information it contains to answer the questions.

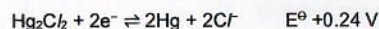
- (a) (i) Suggest why the selective oxidation of primary alcohols to aldehydes is challenging. [1]
 (ii) State the reagents and conditions commonly used for the conversion of primary alcohols to aldehydes and suggest why this method is not green. [2]
 (b) TEMPO is an extremely stable radical that can be stored for years. With reference to its structure and electronic distribution, give two reasons why this is so. [2]
 (c) The mole percentage of a catalyst is defined by:

$$\text{mol \%} = \frac{n_{\text{catalyst}}}{n_{\text{catalyst}} + n_{\text{limiting reagent}}} \times 100\%$$

An industrial oxidation of 10 kg of benzyl alcohol to benzaldehyde was carried out via Process 1. Calculate the mass of TEMPO required for the reaction. [2]

- (d) Use of the Data Booklet is relevant to this question.

The saturated calomel electrode (SCE) is another type of reference electrode based on the equilibrium between mercury(I) chloride and mercury. It has a standard electrode potential of +0.24 V.

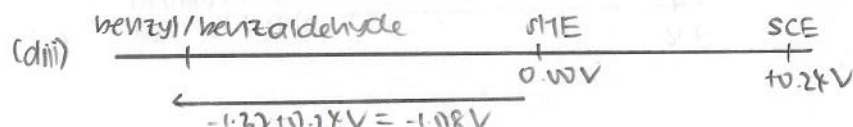


- (i) Suggest why there is a need for other reference electrodes besides the standard hydrogen electrode (SHE). [1]

The electrode potential of a benzaldehyde/benzyl alcohol half cell relative to SCE was measured to be -1.32 V.

- (ii) Write a balanced half equation for the standard electrode potential of the benzaldehyde/benzyl alcohol half cell. [1]
 (iii) Calculate the standard electrode potential of the benzaldehyde/benzyl alcohol half cell. [1]
 (iv) Hence, calculate the standard Gibbs free energy change of equation 1. [2]
 (e) (i) State the oxidation state of nitrogen in nitrosonium cation A and hydroxylamine E. [1]
 (ii) Draw a mechanism for Steps 2 and 3 of process 2. Identify the zwitterionic intermediate D. [4]
 (f) The atom economy of a reaction is defined as the relative molecular mass of the desired product divided by the sum of the relative molecular masses of all substances produced in the stoichiometric equation, expressed as a percentage.

-1.32V wrt SCE [Turn Over]



- (i) Using benzaldehyde as the desired product, calculate the atom economy of process 1 and 2. [1]
 (ii) Atom economy is one of the measures of the extent of Green Chemistry. However, there are limitations to it. Suggest one limitation of atom economy. [1]
 (g) Discuss whether process 1 or 2 is better from the perspective of Green Chemistry. Give a reason to support your answer with reference to relevant data. [1]

[Total: 20]

- 2 Nitrogen molecule and carbon monoxide molecule are isoelectronic.

The molecular orbital diagram showing the valence shell for carbon monoxide is shown in Fig. 2.1.

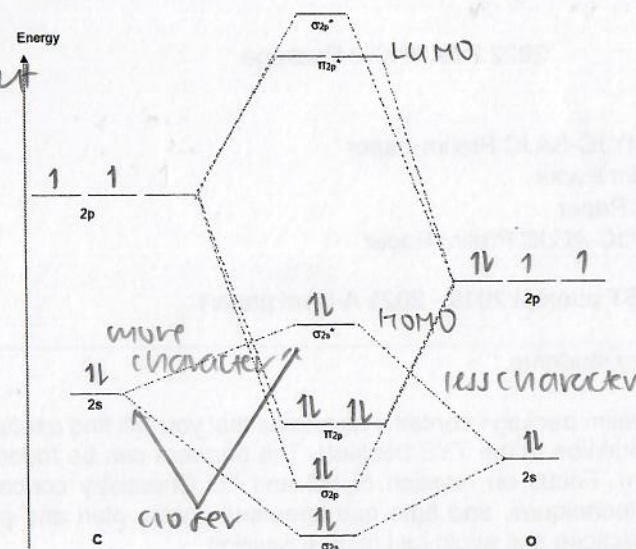


Fig. 2.1

- (a) (i) Explain why the atomic orbitals of oxygen are lower in energy than those of carbon. [2]
 (ii) On Fig. 2.1, populate the orbitals with the appropriate number of electrons. Hence, explain why carbon monoxide has a high bond energy of 1072 kJ mol⁻¹. [2]
 (iii) Reactions, such as complex formation, can be explained by using the HOMO and LUMO.

Carbon monoxide can bind to Fe²⁺ in haemoglobin and the binding is done via the donation of a lone pair of electrons on the carbon atom, instead of that on oxygen atom. With reference to the molecular orbital diagram of carbon monoxide, explain why this is so. [2]

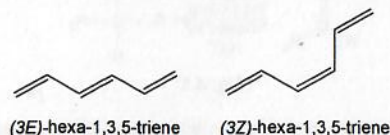
- carbon lone pair from sigma*_2p, which is closer in energy to carbon and so will donate the lone pair of electrons

- (b) Table 2.1 lists the ultraviolet (UV) wavelength of maximum absorbance for some organic compounds.

Table 2.1	
compound	Wavelength of maximum absorbance, λ_{max} / nm
$\text{CH}_2=\text{CH}_2$	171
Hexa-1,3,5-triene	258

- (i) Outline the principles of UV-vis spectroscopy of organic molecules. [2]
- (ii) By considering orbitals of π symmetry, draw the HOMO and LUMO of hexa-1,3,5-triene. [2]
- (iii) In the UV-vis analysis of hexa-1,3,5-triene, benzene should not be used as a solvent even though there is no reaction between these two compounds. Suggest a reason for this.
will overlap and mask signal [1]
- (iv) A sample of hexa-1,3,5-triene was analysed using UV-vis spectroscopy. Its concentration was $1.8 \times 10^{-8} \text{ mol dm}^{-3}$ and the pathlength of the cuvette used was 1 cm. Given that 20% of the UV light was transmitted, calculate the molar absorptivity of hexa-1,3,5-triene at 258 nm. Include unit if any.
Beer Lambert Law: $A = \epsilon c l$ [2]

- (v) Hexa-1,3,5-triene can exist as either the E or Z isomers.
w/ r/ c $\epsilon = 1000 \text{ dm}^2 \text{ mol}^{-1} \text{ cm}^{-1}$



The conversion of one isomer to the other can only happen when light of wavelength 258 nm is shone. Explain this observation. [2]

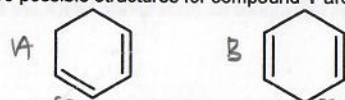
- (vi) (3E)-hexa-1,3,5-triene can undergo a reaction to form compound X, which gives rise to the following infra-red spectrum in Table 2.2.
energy enough to overcome barrier to rotation $\pi \rightarrow \pi^*$ and π bond is broken - can rotate about bond

no sp² C-H stretch
no alkene C=C stretch
no alkene C-H stretch

Table 2.2	
wavenumber / cm^{-1}	description
3300 cm^{-1}	Strong and broad peak
2900 cm^{-1}	Strong peak
1100 cm^{-1}	Strong peak

Suggest the reagents and conditions to convert (3E)-hexa-1,3,5-triene into X.
C-O stretch

- (c) (i) (3E)-hexa-1,3,5-triene can undergo a reaction to form a single product known as compound Y. The two possible structures for compound Y are as shown.



Describe a chemical test, with appropriate observations, to distinguish between these two structures.
effluence no effluence
oxidative cleavage + lime water test for CO_2 [2]

- (ii) State and explain the difference in the UV-vis spectra for the two possible structures of Y. [2]

A: conjugated π system UV-vis absorption

max intensity

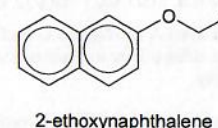
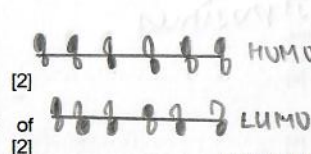
B: Isolated π bond UV-vis absorption, λ_{max} (another difference if required)

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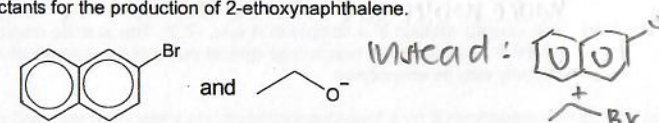
[Turn Over

- 3 The Williamson ether synthesis is widely used to produce ethers. It is a nucleophilic substitution reaction in which an alkoxide ion, R-O^- , replaces a halide ion from an alkyl halide.

2-ethoxynaphthalene is a commonly used ether in perfumery for its odour of orange blossoms.



A student proposed two reactants for the production of 2-ethoxynaphthalene.



However, the teacher commented that there will be no reaction between these two reactants.

- (a) (i) Suggest two reasons why these two reactants will not react.
Dipolar double bond character [2]
steric hindrance
- (ii) Hence, suggest two reactants that can be used to produce 2-ethoxynaphthalene.
1-ethoxynaphthalene [1]

When A is heated with $(\text{CH}_3)_3\text{COK}$, an E2 reaction takes place to form alkenes C and D but not B.

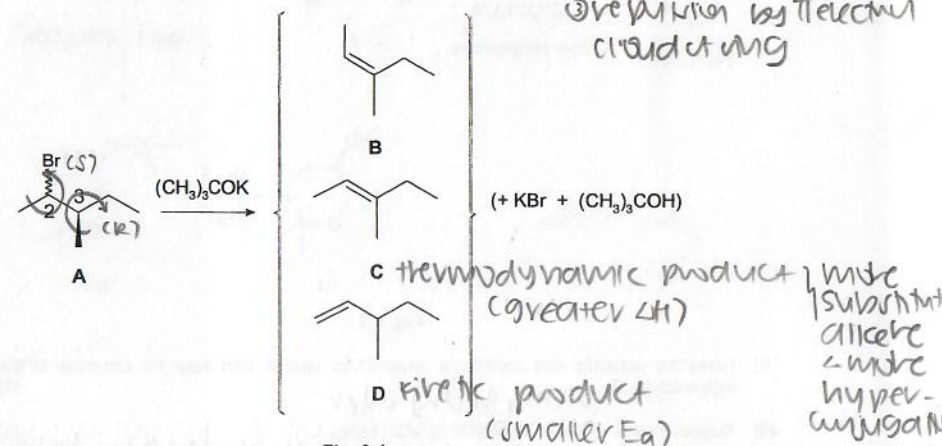
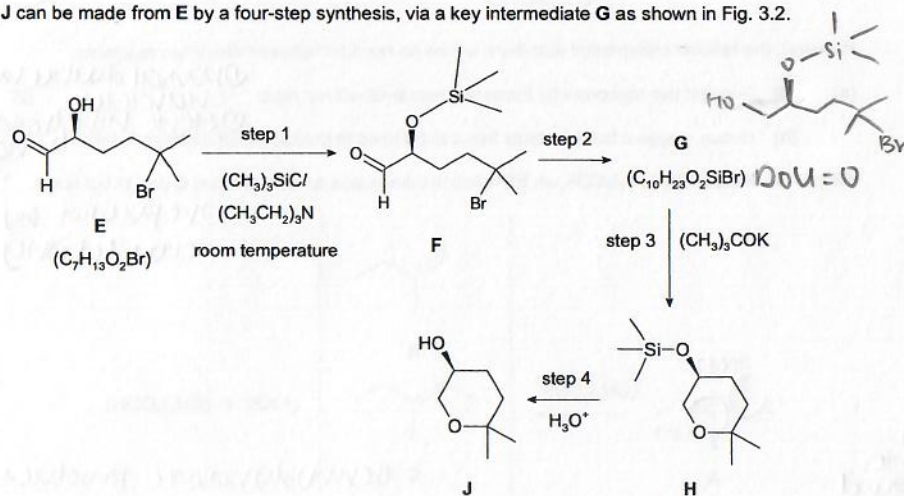


Fig. 3.1

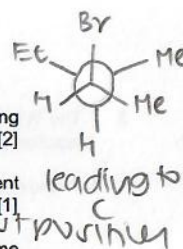
Table 3.1 shows the percentages of alkenes B, C and D at different temperatures.

experiment		percentages of alkenes		
		B	C	D
1	room temperature	0	40	60
2	heat	0	65	35

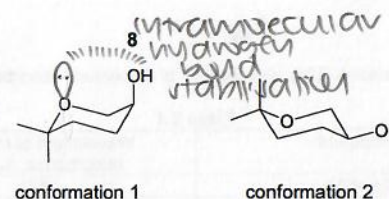
- (b) (i) Draw the Newman projections through carbon 2 and 3 that lead to C. Hence, considering the percentage yields, determine the *R/S* configuration of carbon 2 in A. [2]
- (ii) Suggest an explanation for the higher percentage yield of alkene D than C in experiment 1. *Less steric hindrance when abstracting at 1st position* [1]
- (iii) Sketch the energy profile diagrams from A to alkenes C and D respectively on the same axes. Hence, with reference to the differences in their enthalpy changes and activation energies, explain your diagrams fully. [4]
- (iv) Using the data in Table 3.1 and your answer in (b)(iii), explain why the percentages in experiment 2 differs from experiment 1. *More molecular movement sufficient E_k to overcome E_a for reaction to proceed* [1]
- (v) The specific rotation of a sample of A was -7.3° . The specific rotation of optically pure A is $+66.9^\circ$. Calculate the percentage optical purity of this sample of A, assuming the only impurity was its enantiomer. [1]
- (c) J can be made from E by a four-step synthesis, via a key intermediate G as shown in Fig. 3.2.



- (i) State the reagents and conditions required for step 2, and draw the structure of the intermediate G. *silane in dry ether* [2]
- (ii) Suggest the purpose of using $(\text{CH}_3)_3\text{SiCl}$ in step 1. *protecting group to prevent alcohol from being dehydrated* [1]
- (iii) Suggest why NaOH cannot be used in step 3. *act as Nu and attacked carbon to form alkoxide to undergo $\text{S}_\text{N}2$ reaction* [1]
- (iv) Draw the mechanism for step 3 in Fig. 3.2. [2]
- (v) Two possible chair conformations of J are shown below.



* Since yield of B=0, it must have an R-configuration at carbon-2



Explain, with an aid of a suitable diagram, why conformation 1 is more stable than conformation 2. [2]

intramolecular H-bonds stabilize the molecule much more than the stabilisation from reduced 1,3-diaxial interactions between OH and H.

Section B
Answer two questions in this section.

- 4 Most of the chemistry of magnesium relates to its ionic compounds. However, magnesium does form an important group of covalent compounds which are known as Grignard reagents.

A typical example of the use of a Grignard reagent is the use of alkyl magnesium bromide, RMgBr , with a carbonyl compound, via a nucleophilic addition reaction, to give an alcohol as shown in Fig. 4.1.

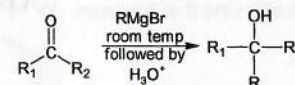
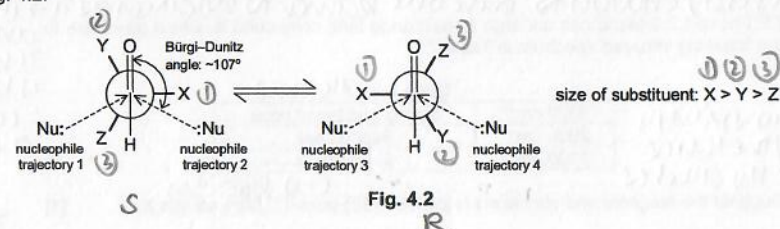


Fig. 4.1

In 1969, the Felkin-Anh model was proposed. In this model, the most stable conformation of an aldehyde has the largest group perpendicular to the carbonyl group, staggered substituents with no eclipsing interactions and large substituents as far apart from one another as possible, as shown in Fig. 4.2.



When any nucleophile attacks the carbonyl carbon, it will do so at the Bürgi-Dunitz angle (about 107° from the $\text{C}=\text{O}$ bond). The 4 possible trajectories for nucleophilic attack are shown in Fig. 4.2.

A undergoes a series of reactions as shown in Fig. 4.3.

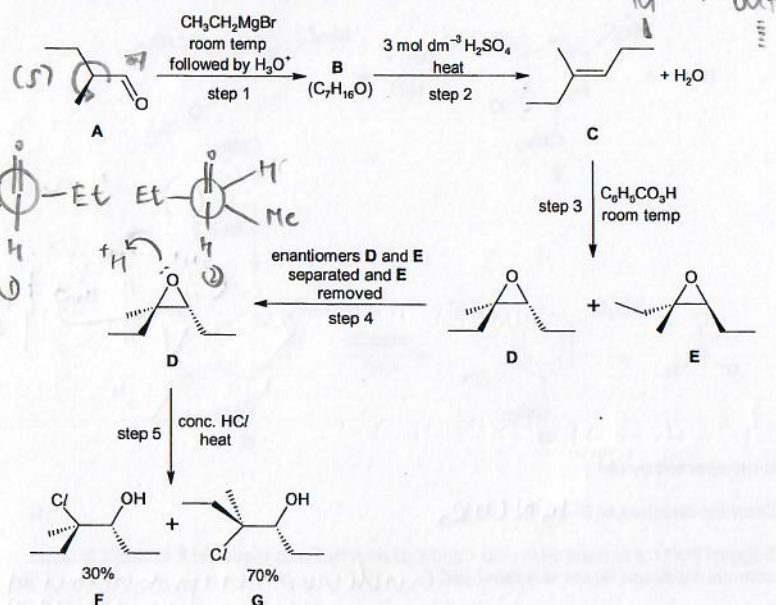


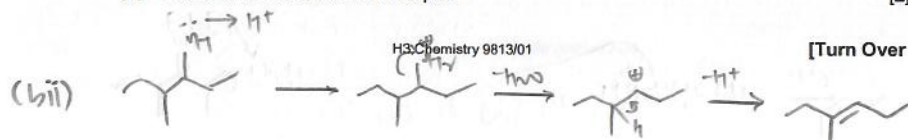
Fig. 4.3

Step 2 was further investigated by modifying the concentration of H_2SO_4 and temperature. The results are shown in Table 4.1.

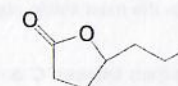
Table 4.1

temp / $^{\circ}\text{C}$	$[\text{H}_2\text{SO}_4] / \text{mol dm}^{-3}$	relative rate	% yield of C in step 2
25	3	1.00	less than 40%
170	3	1.75	more than 90%
170	5	1.74	more than 90%

- (a) (i) With reference to Fig. 4.2, explain why trajectory 1 is the most favourable for nucleophilic addition. *Least steric hindrance encountered as other trajectories are cluttered by larger substituents* [1]
- (ii) Draw the two most stable Newman projections of A, viewing through from the aldehyde carbon. [1]
- (iii) Hence, draw the structure of B, the major product in step 1. You are expected to show all stereochemistry clearly. [2]
- (b) (i) With reference to Table 4.1, suggest and explain the overall order of reaction in step 2. Hence, identify the mechanism for step 2. *Elimination E1* [2]
- (ii) Describe the mechanism in step 2. [2]

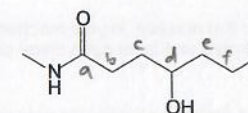


- (iii) Using relevant data from the *Data Booklet*, calculate the enthalpy change of reaction in step 2. [1]
- (iv) Hence, using the concept of Gibbs free energy, explain fully why step 2 of the reaction in Fig. 4.3 gives a higher percentage yield of C at higher temperature as shown in Table 4.1. *more products -> 1570 and 1710 cm-1* [2]
- (c) (i) Name the type of reaction that C undergoes in step 3. *oxidation* [1]
- (ii) D and E are vaporised and the bond energy of C-O measured. Apart from bond energy values being averaged values, suggest why the bond energy of C-O in D and E are significantly lower than the value stated in the *Data Booklet*. *more strain - nonideal between bond pair* [1]
- (d) (i) Suggest the purpose of adding an acid in step 5. *to activate nucleophile* [1]
- (ii) Suggest why D undergoes $\text{S}_{\text{N}}1$ in step 5 but not $\text{S}_{\text{N}}2$. *Bulky group at tertiary carbon prevents nucleophilic attack* [2]
- (iii) Describe the mechanism in step 5. *more acid increases nucleophilicity* [2]
- (iv) Explain why the percentage yield of G is higher than F in step 5. *more strain - nonideal between bond pair and steric hindrance more force - favour backside attack* [2]
- 5 (a) Lactones are cyclic esters and they can react with amines. Compound A is an example of a lactone.



Compound A

- (i) Compound A reacted with an amine to form Compound B.

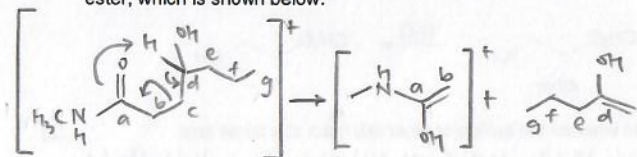


Compound B

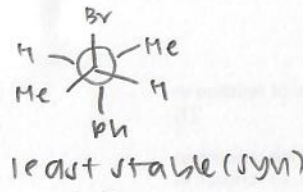
State the reagent needed to form B. *CH_3NH_2*

[1]

- (ii) Fragmentation of molecular ions in the mass spectrometer is sometimes accompanied by rearrangement. An example is the McLafferty rearrangement of the molecular ion of an ester, which is shown below.

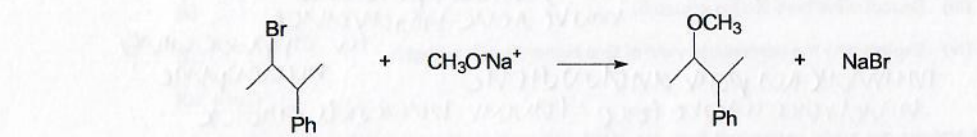


most stable (anti-staggered)



least stable (syn)

(b) Compound C, (2R,3S)-2-bromo-3-phenylbutane was treated with sodium methoxide in methanol according to the following equation and the reaction produced a mixture of organic products.



-C₆H₅ is represented by Ph

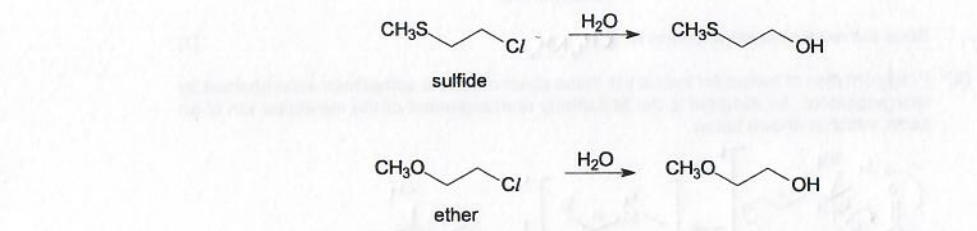
(i) Draw Newman projections to show the most stable staggered and least stable eclipsed conformations of C. [2]

(ii) A possible mechanism for the reaction between C and sodium methoxide in methanol proceeds via neighbouring group participation.

The first step of this mechanism involves the phenyl group acting as a nucleophile via S_N2. The reactive intermediate formed then reacts with methoxide ion and regenerates the benzene ring.

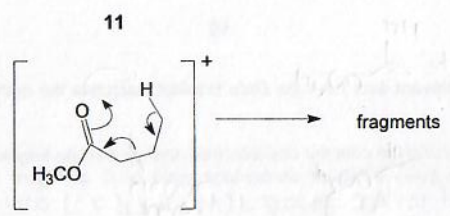
Suggest the mechanism for this reaction. In your mechanism, include the stereochemistry for the organic products formed and label each chiral centre using the R, S convention. [3]

(iii) Both sulfides and ethers can undergo a similar mechanism involving a neighbouring group participation.



Predict and explain whether the sulfide or ether will react at a faster rate. [2]

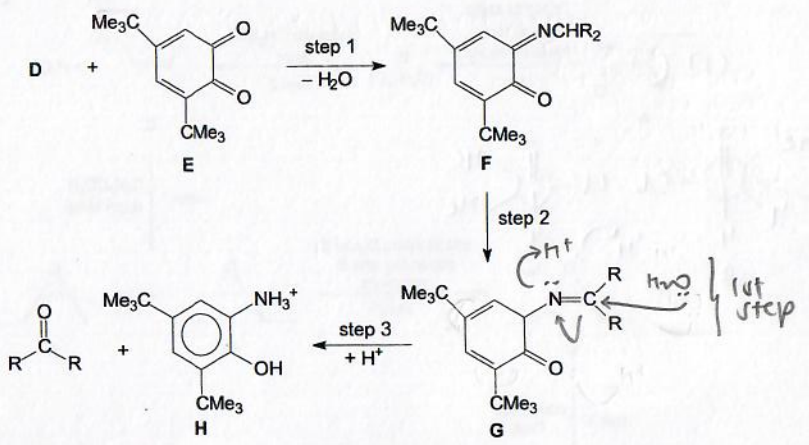
faster. lone pair more available for donation as S is less EN than O. [Turn Over]



Draw the fragments formed when molecular ion of compound B undergoes McLafferty rearrangement. [1]

(iii) Hence, identify the ions responsible for the peaks which occur at m/z 30, 43, 73 and 86 in the mass spectrum of B. [2]

(c) Primary amine D is oxidised to a ketone by ortho-quinone E, via intermediates F and G.

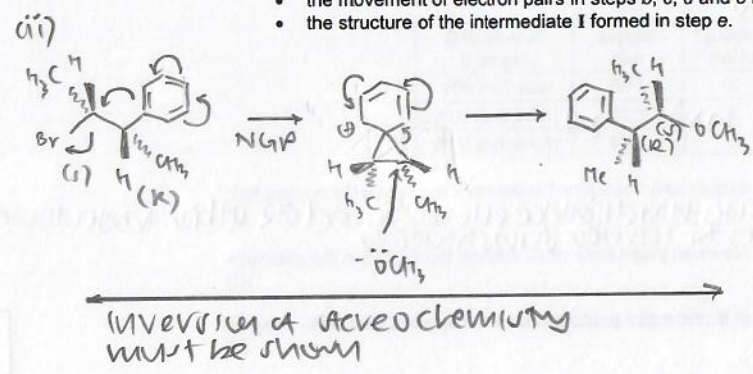


-CH₃ is represented by Me

- (i) Draw the structure of D. H_2NCHR_2 [1]
- (ii) Suggest the type of reaction in step 1 and explain why the formation of F in step 1 is faster near neutrality but slower at a lower pH. *condensation + protonation of N make lone pair unavailable to contribute a bond* [2]
- (iii) State and explain the relationship between F and G. [1]
- (iv) G undergoes acidic hydrolysis in step 3 to form H. [1]

Complete Fig. 5.1 to suggest a mechanism for this reaction. In Fig. 5.1, you will need to show:

- the movement of electron pairs in steps b, c, d' and e by using curly arrows
- the structure of the intermediate I formed in step e. [3]



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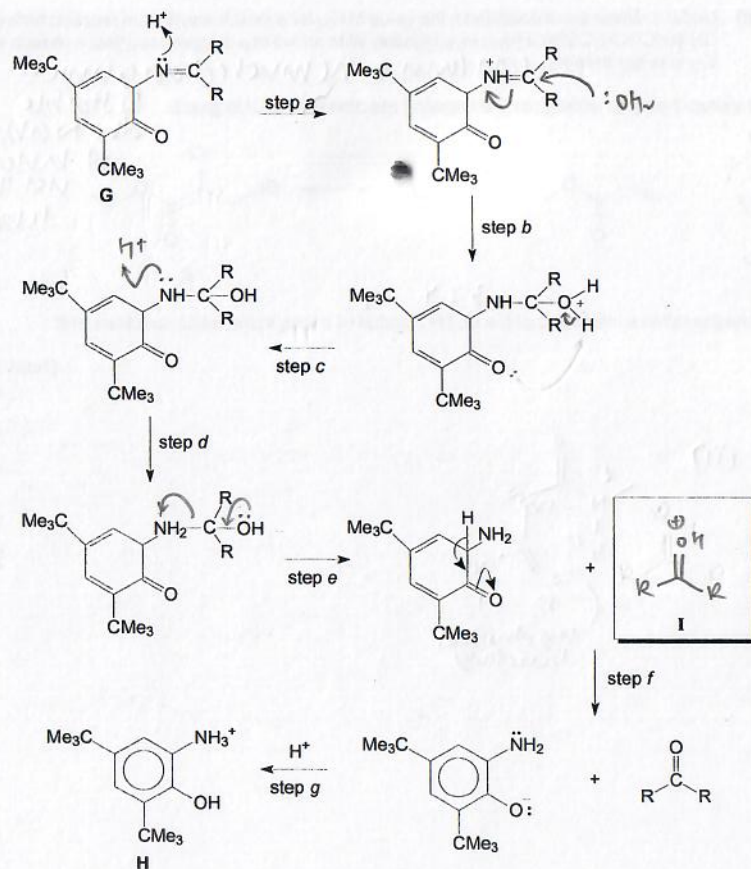


Fig. 5.1

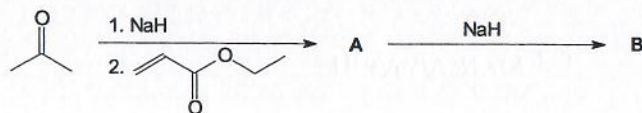
(v) Explain why protonation of the nitrogen atom is necessary in step a of the mechanism in (c)(iv). *make C more electrophilic and susceptible to nucleophilic attack*

(vi) Suggest two key differences in the chemical shift of 1H NMR spectrum between E and H. *1) N-H labile proton signal, 2) O-H labile proton signal*

[1]

[Total: 20]

6 (a) Propanone undergoes the following sequence of reactions.



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[Turn Over]

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The infra-red and 1H NMR spectra of compound A are shown in Fig. 6.1 and 6.2.

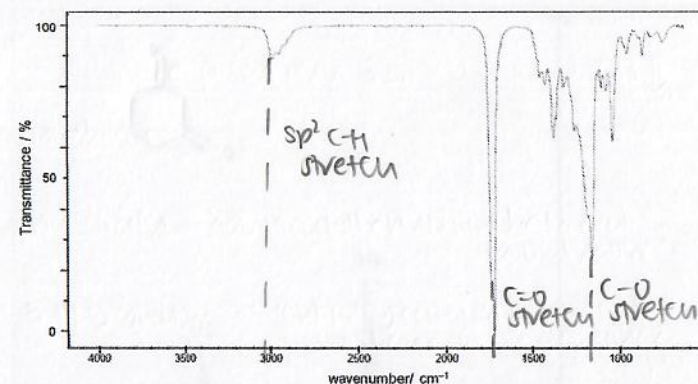
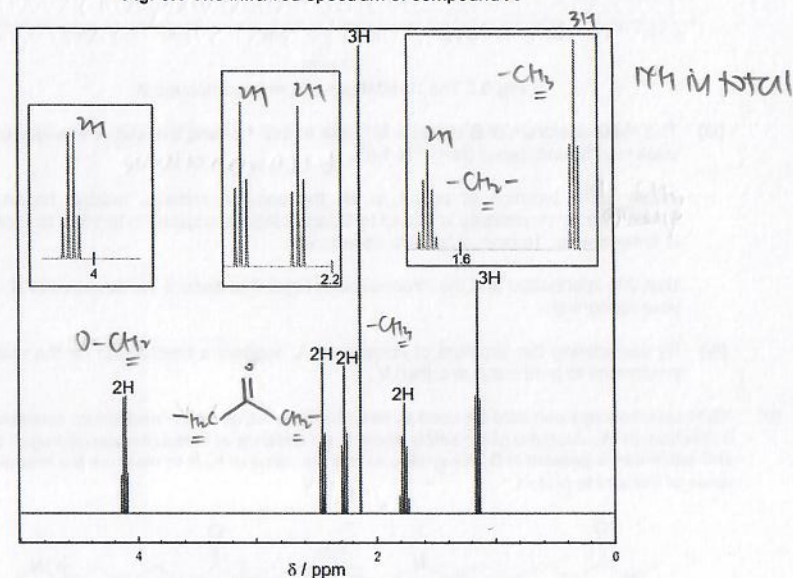


Fig. 6.1 The infra-red spectrum of compound A

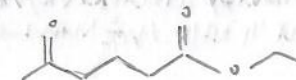
Fig. 6.2 The 1H NMR spectrum of compound A

(i) Outline the basic principles of 1H NMR. [3]

(ii) The mass spectrum of A shows major peaks at m/z values of 43 and 113, and a M^+ peak at m/z 158. *molecular formula: $C_8H_{14}O_3$*

Use this information and the information in Fig. 6.1 and 6.2 to deduce the structure of A. Explain your reasoning. [5]

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The ^1H NMR spectra of compound B is shown in Fig. 6.3.

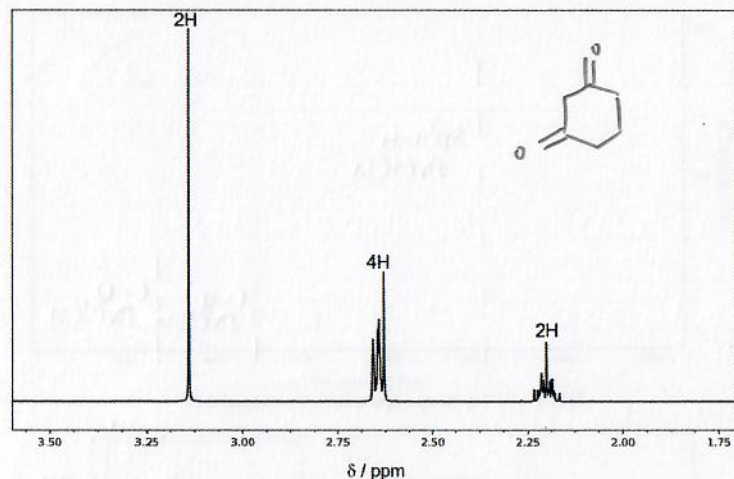


Fig 6.3 The ^1H NMR spectrum of compound B

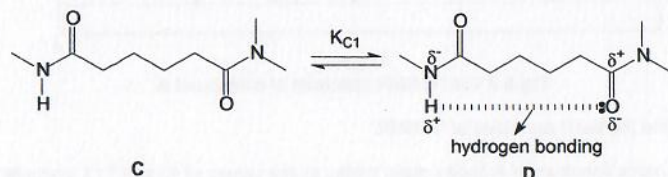
- (iii) The mass spectrum of B shows a M^+ peak at m/z 112 and the relative intensity of the M^+ peak is 27% and that of $(M+1)^+$ is 1.8%. *6 carbon atoms*

When liquid bromine is added to B, the solution remains reddish brown. When 2,4-dinitrophenylhydrazine is added to B, an orange precipitate is formed. No precipitate is formed when Tollens' reagent is added to B.

Use this information and the information in Fig 6.3 to deduce the structure of B. Explain your reasoning. [5]

- (iv) By considering the structure of compound A, suggest a mechanism for the reaction of propanone to produce A and then B. [3]

- (b) NMR spectroscopy can also be used to determine the value of the equilibrium constant, K_c , of a reaction. In the example given below, there is an absence of intramolecular hydrogen bonding in C while this is present in D. A key step to find the value of K_c is to measure the chemical shift value of the amide proton.

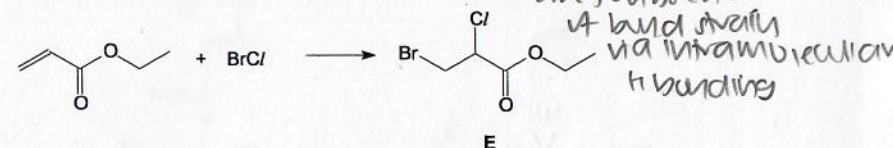


- (i) Suggest a reason for the use of CDCl_3 as a solvent to record the NMR spectrum of the above compound. [1]

No appreciable signal from deuterated solvent in ^1H NMR spectra which may mask signals

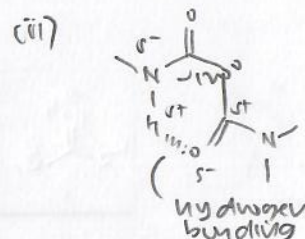
- (ii) Under a similar set of conditions, the value of K_{C2} for a similar equilibrium reaction between $\text{CH}_3\text{NHCOCH}_2\text{CON}(\text{CH}_3)_2$ is investigated. With an aid of a diagram, suggest a reason why K_{C2} is larger than K_{C1} . *can form 6-membered ring which is stable due to absence of hindrance via intramolecular H bonding* [2]

- (c) Ethylprop-2-enoate undergoes the following reaction with BrCl to give E.



State the relative intensities of the M, M+2 and M+4 peaks in the mass spectrum of E. [1]

[Total: 20]



*① HNC=C
② HNC=O
③ HNC=O*

Information for Question 1

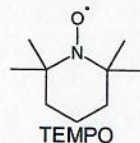
Efficiency in organic synthesis has traditionally been defined by chemical yield. However, in recent times, there has been a shift towards Green Chemistry – where efficiency is defined in terms of minimising waste and avoiding toxic or hazardous materials.

Here are some principles that demonstrate the concept of Green Chemistry:

1. **Maximise atom economy:** Design syntheses so that the final product contains the maximum proportion of the starting materials, wasting few or no atoms.
2. **Design less hazardous chemical syntheses:** Design syntheses to use and generate substances with little or no toxicity to either humans or the environment.
3. **Use safer solvents and reaction conditions:** Avoid using solvents or separation agents. If these chemicals must be used, safer ones should be considered.

Aldehydes are important intermediates in the synthesis of pharmaceuticals, agrochemicals and fine chemicals. They are often prepared via alcohol oxidation. Selective oxidation of primary alcohols to aldehydes is one of the long standing problems of organic chemistry.

Two possible processes for the selective oxidation of alcohols to aldehydes involving the green organocatalyst TEMPO (2,2,6,6-tetramethylpiperidinyloxy) are presented below.

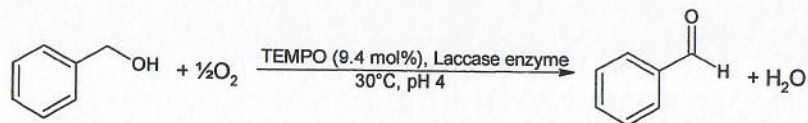


Abstract 1 (M. Fabbrini et al., *Tetrahedron Lett.*, 2011, 42, 7551)

Process 1

This method involves using Laccase, an enzyme secreted by white rot fungi, in combination with TEMPO, to catalyse the oxidation of primary benzylic alcohols using atmospheric oxygen.

Equation 1

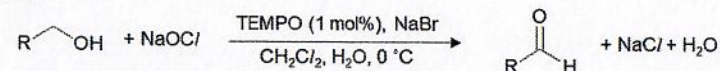


Abstract 2 (R. Sheldon et al, *Chem. Soc. Rev.*, 2012, 41, 1437)

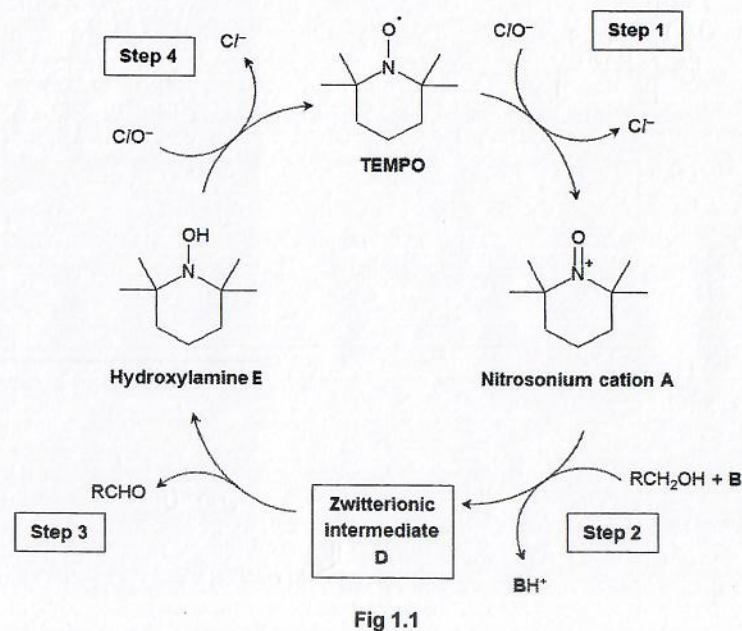
Process 2

This method, involving hypochlorite (household bleach) as the terminal oxidant, is suitable for the oxidation of a broad range of alcohols.

Equation 2



The chemical cycle of Equation 2 is shown in Fig 1.1.



- Step 1:** TEMPO undergoes oxidation with ClO^- to form the reactive nitrosonium cation A.
- Step 2:** The alcohol is deprotonated by a base B. The nitrosonium cation A reacts with the deprotonated alcohol to give a zwitterionic intermediate D.
- Step 3:** Zwitterionic intermediate D undergoes a single-step reaction, forming the aldehyde product and hydroxylamine E.
- Step 4:** Hydroxylamine E undergoes oxidation with ClO^- to regenerate TEMPO.

